MILITARY SPECIFICATION

FLASH UNITS, PHOTOGRAPHIC REPEATING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. - This specification covers the requirements for two types of portable, light-weight, electronic flash units for photographic use.

1.1 CLASSIFICATION

1.2.1 Types. - This specification covers electronic flash units of the following types:

Type I - Medium performance unit having a minimum light output of 1200 Beam Candle Power Seconds (BCPS) designated as LE-25( ).

Type II- High performance unit having a minimum light output of 2400 (BCPS) designated as LE-26( ).

The complete type designator will be provided by the requiring activity upon application by the contractor (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 SPECIFICATIONS AND STANDARDS. - The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-571	Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and solid Form.

Military

MIL-P116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, AN Type
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

FSC 6760

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MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-M-9868	Microfilming of Engineering Documents, 35mm, Requirements for
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid "Rosin Base"
MIL-T-16910	Interference Measurement, Radio, Methods and Limits, 14 Kilocycles to 1000 Megacycles
MIL-B-55130	Batteries, Storage, Sealed, Nickel-Cadmium
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-155	Joint Photographic Type Designation System
MIL-STD-169	Extreme Temperature Cycle
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MS-75031	Battery, Storage, BB-412( )/U
MS-75032	Battery, Storage, BB-416( )/U
MS-91184	Connector, Plug, Electrical Power, Non-Polarized, Parallel Blade, Type UP120M

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 OTHER PUBLICATIONS. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 Screw Thread Standards for Federal Services

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

AMERICAN STANDARDS ASSOCIATION

ASA Standard PH3.6 Tripod Connections for American Cameras, 1/4 inch-20 thread

ASA Standard PH3.30 Camera Accessory Shoes, Dimensions for

ASA Standard PH3.40 Photographic Electronic Flash Equipment, Method of Measuring and Designating the Performance of

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, N. Y.). (Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 DESCRIPTION. - Flash Units, Photographic, Repeating, Type I and Type II shall be designed and constructed to provide reliable, high quality, rugged, portable and light-weight repeating electronic flash types of light sources of daylight quality. The flash units shall be capable of being used with any camera equipped with zero time delay, X type, synchronization contacts for flash photography. Each flash unit shall be capable of operation from a re-chargeable battery and from an external 115 volt, 50-60 cycle, single phase, alternating current (ac) power source. Each flash unit shall have a built-in battery charger. The flash units shall be so constructed that parts are easily and readily accessible and removable with a minimum of effort for servicing. Each flash unit shall consist of the following major components: Power Supply, Battery, Light Unit(s), External Power Cable and Light Unit Mounting Accessories.

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3.2 DATA. - No data is required by this specification or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.3 APPLICABILITY OF REQUIREMENTS. - Unless otherwise specified herein, all the requirements of this specification shall apply to both Type I and Type II flash units.

3.4 PREPRODUCTION SAMPLE. - Unless otherwise specified (see 6.2), one Type I or Type II Flash Unit, whichever is ordered, shall be manufactured using the same methods and processes as will be used for quantity production and shall be submitted as the preproduction sample for acceptance. Approval of the preproduction sample by the requiring activity shall not relieve the contractor of his obligation to supply equipment conforming to this specification.

3.5 MATERIAL. - Material shall be as specified herein. Materials not definitely specified shall be of good commercial quality suitable for the purpose. Material shall be free from all defects and imperfections that might affect the serviceability or performance of the finished product.

3.5.1 Plastic. - Plastic material, if used for the cases, shall be a tough, flexible, high impact type capable of withstanding long periods of rough usage and shall not be adversely affected by environmental extremes of temperature and humidity.

3.5.2 Metal. - Metal, if used for the cases, shall be any aluminum alloy suitable to meet the requirements of this specification.

3.5.3 Fungus Resistance.- Materials which are not nutrients for fungus growth shall be used to the maximum extent possible. All other materials shall be suitably finished or treated to resist the growth of fungi.

3.6 RELIABILITY. - The flash unit shall comply with the requirements of this specification when subjected to the component reliability test in Section 4.

3.7 INTERCHANGEABILITY. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment, unless otherwise specified herein. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

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3.8 CONSTRUCTION

3.8.1 Riveting. - Riveting shall not be used for mounting parts such as capacitors, resistors, and transformers. Rivets shall be used where practicable in preference to other means of securing aluminum structural parts if the joint is subjected to dynamic tension loads when the flash unit is exposed to specified service conditions and if the parts do not require removal when the flash unit is serviced.

3.8.2 Solder. - Solder shall conform to Federal Specification QQ-S-571. For soldering electrical connections, composition SN-60, Type AR or S, shall be used for general purposes; and composition SN-62 or SN-63, Type AR or S, for special purposes. Where Type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.8.2.1 Acid or Acid Salts. - No acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.8.2.2 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles or other foreign material remaining in the items, flux being spread over a large area, or damage to the items. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.8.3 Screw Threads. - Screw threads employed in fabrication of the flash unit shall conform to the National Bureau of Standards Handbook H-28 and be commercial standard sizes.

3.8.4 Cleaning. - After fabrication, parts shall be cleaned in accordance with good commercial practice. Cleaning procedures shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels etc. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.9 DIMENSIONS. - The approximate dimensions of the power supplies and light units shall be as follows:

	Width	Depth	Height
Type I Power Supply	6 inches	3 inches	6 inches
Type II Power Supply	9 inches	3 inches	7 inches

	Diameter	Depth
Type I Light Unit	5 1/2 inches	4 1/2 inches
Type II Light Unit	5 1/2 inches	4 1/2 inches

3.10 WEIGHT. - The approximate weight of the power supplies (with batteries) and the light units shall be as follows:

Type I Power Supply	3 1/2 lbs.
Type II Power Supply	7 lbs.
Type I Light Unit	3/4 lbs.
Type II Light Unit	3/4 lbs.

3.11 SERVICE CONDITIONS. - The flash unit shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered as compliance with the requirements.

3.11.1 Temperature - Operating. - Ambient temperature in the range of plus 125 to 0 degrees F with a relative humidity up to 95 percent, with performance as follows:

Temperature	Performance
Plus 77 plus or minus 5 degrees F.	As specified in 3.12.1, 3.12.2, 3.12.5, and 3.12.6.
Plus 125 degrees F.	As specified in 3.12.1, 3.12.2, 3.12.5, and 3.12.6 except there shall be no more than 10 percent light loss.
0 degrees F.	As specified in 3.12.1, 3.12.2, 3.12.5, and 3.12.6 except there shall be no more than 50 percent light loss and the recycling time shall not exceed 15 seconds.

3.11.2 Temperature - Non-Operating. - Exposure in the range of plus 160 degrees F with a relative humidity up to 30 percent, to minus 65 degrees F with relative humidity up to 95 percent; however, exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours at any one time.

3.11.3 Relative Humidity.- The flash unit shall be capable of withstanding high relative humidity for long periods of time and shall pass the Humidity Test of Section 4.

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3.11.4 Vibration. - The flash unit shall withstand without damage the effects of vibration to pass the Vibration Test of Section 4.

3.11.5 Impact. - No physical damage except for surface abrasions, small dents, and breakage of external knobs, etc., that can be replaced without disassembly of the unit when subjected to the Impact Test in Section 4.

3.11.6 Elevation. - Operation at elevations up to 10,000 feet and non-operating elevations up to 50,000 feet above sea level.

3.12 PERFORMANCE. -

3.12.1 Light Output. - The minimum light output when measure in accordance with the procedures described in American Standard PH3.40 shall be as follows:

Type I Flash Unit - 1,200 Beam Candle Power Seconds (BCPS)
Type II Flash Unit - 2,400 BCPS

3.12.2 Recycle Time.- The recycle time shall not exceed 10 seconds when measured in accordance with the procedures described in American Standard PH3.40.

3.12.3 Reflector Coverage. - The reflector shall have an angle of coverage of 60 degrees when measured in both the vertical and horizontal planes in accordance with the procedures described in American Standard PH3.40.

3.12.4 Synchronization. - The flash unit shall be capable of being flashed in synchronization with any camera equipped with X type, zero delay, synchronization contacts for flash photography.

3.12.5 Flash Duration. - The flash duration shall not be shorter than 1/2500 second when measured in accordance with the procedures described in American Standard PH3.40.

3.12.6 Number of Flashes Per Battery Charge. - The number of flashes from a fully charged battery, when the flash unit is operated on BAT and flashed every 15 seconds, shall not be less than 100 for the Type I Flash Unit and not less than 200 for the Type II Flash Unit. The light output of any of these flashes shall not be less than that specified herein.

3.12.7 Battery Charging Rate. - The battery charging rate of the power supply shall be capable of charging a completely discharged battery in 14 hours or less and the battery shall not be damaged if left on charge for as long as 24 hours.

3.12.8 Operating Life. - The flash unit shall be capable of supplying a minimum of 10,000 flashes without any replacement or adjustment of any parts or components. The light output shall not fall below 90 percent of that specified in this specification. There shall be no pitting, burning, or any damage to the internal shutter contacts of any camera used in this test which will cause any malfunctions.

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3.13 RADIO INTERFERENCE SUPPRESSION. - Unless otherwise specified in the contract or order (see 6.2), the flash unit shall meet the emanation limits of MIL-I-16910 relative to conducted or radiated electrical disturbances, not including transients which occur when the lamp is flashed, which may interfere with the operation of electrical or electronic equipment. Application of suppression components and techniques shall conform to that specification.

3.14 POWER SUPPLY. -

3.14.1 Case. - The power supply case shall be constructed of tough, flexible, high-impact plastic material or light-weight metal. The material selected shall be capable of withstanding long periods of rough usage and shall not be adversely affected by environmental extremes of temperature and humidity. The case may be of a sectionalized or unitized construction and shall be provided with a suitable removable cover. The cover and case when assembled shall prevent the entrance of water into the interior of the case.

3.14.2 Carrying Strap.- An adjustable one inch wide web or woven nylon strap with a two inch wide protective pad made of polyurethane shall be provided with each power supply. The length of the strap shall be such that individuals ranging from five feet to six feet three inches in height may carry the power supply with comfort. The strap shall be fastened to the power supply in such a manner that it can be easily removed and cannot become unfastened during use. The color of the strap shall be a medium gray unless otherwise specified by the requiring activity in the contract or order (see 6.2).

3.14.3 Power Control Switch. - A power control switch shall be provided. The switch shall be a two position type and shall be of such size and shape as to permit operation by a gloved hand. The switch positions shall be marked or labeled ON-OFF. When the switch is placed in the OFF position, all functions shall be rendered inoperative and when placed in the ON position, only the single function selected by the selector switch shall be operative.

3.14.4 Selector Switch. - A selector switch shall be provided to select the mode of operation required. The switch shall be a three position switch and shall be of such size and shape as to permit operation by a gloved hand. The positions of the switch shall be marked AC -BAT - CHG. The switch when placed in the AC position, shall permit operation of the unit only from an external ac power source. When the switch is placed in the BAT position, the unit shall operate only from the power of its self contained battery. The CHG position of the switch, shall permit only charging of the battery. The switch shall be capable of changing the mode of operation without causing any damage to the components of the electrical circuits or danger to operating personnel when the power control switch is in the ON position.

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3.14.5 External Power Cable Connector. - One recessed type male connector shall be provided on the power supply for attaching the external power cable. The connector may be a twist lock type.

3.14.6 Light Unit Connector(s). - One female type connector shall be provided on the Type I power supply and two female type connectors, wired in parallel, shall be provided on the Type II power supply for attaching the light unit cable. The connectors may be commercial or AN type as specified in the contract or order. (see 6.2). The AN type, if used, shall conform to the requirements of Military Specification MIL-C-5015. If a commercial type connector is used, it shall be capable of mating with the light unit cable connector with complete safety to the equipment and to operating personnel under all specified conditions. The connectors shall not be interchangeable between the power supplies of the Type I and Type II flash units.

3.15 BATTERY. - A battery shall be provided consisting of five individual, sealed, rechargeable, nickel-cadmium, inter-connected cells encased in a plastic housing and provided with means for making electrical connections.

3.15.1 Individual Cells. - Unless otherwise specified in the contract or order (see 6.2), the individual cells of the battery for use in the Type I power supply shall be type BB-416( )/U and for use in the Type II power supply shall be type BB-412( )/U. The cells shall conform to the requirements of Specification MIL-B-55130, Military Standard MS-75031 and MS-75032 as applicable.

3.15.2 Battery Housing. - The five interconnected cells shall be encased in a plastic housing meeting the requirements of paragraphs 3.4.1 (Case Fit) and 3.4.2 (Container Alkali Absorption Resistance) of Specification MIL-B-55130 and as specified herein.

3.15.3 Battery Leads. - If battery leads are used, the positive and negative poles of the battery shall terminate in leads of sufficient length to connect the battery to the power supply battery terminals. The leads shall have non-interchangeable, quick disconnect terminals and shall be colored red and black for the positive and negative poles respectively.

3.15.4 Useful Life. - The battery shall have a minimum useful life of 100 cycles of charge and discharge. Upon completion of the life test in Section 4, the battery capacity shall not decline below 95 percent of its nominal rated capacity.

3.16 LIGHT UNIT. - Unless otherwise specified in the contract or order (see 6.2), only one light unit shall be provided with each flash unit ordered.

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3.16.1 Housing. - The light unit housing shall be made of the same material used in the fabrication of the power supply case. The housing construction shall provide easy accessibility to the interior parts for maintenance or repair without the use of special tools.

3.16.2 Reflector. - The reflector shall be constructed on non-tarnishing material suitable for the intended purpose. The design of the reflector shall enable the light unit to meet the specified light output and angle of coverage requirements. The reflector shall be protected by a clear, colorless, plastic shield.

3.16.3 Flashtube. - The flashtube shall be a Xenon filled tube designed for the intended application. The flashtube shall be capable of withstanding the maximum power loading for a minimum useful life of 10,000 flashes when flashed at a rate on one flash every 15 seconds.

3.16.4 Cable. - The light unit shall be provided with a coiled cable 30 inches long when compressed and 6 feet long when extended. The cable shall be provided with a connector on one end to mate with the light unit connector on the power supply and shall be strain relieved on the light unit end. The cable shall be capable of carrying the high voltages from the power supply to the light unit with complete safety to operating personnel.

3.16.5 Ready Light. - An indicator light, of the flashing type, shall be provided which shall not start to operate until the high voltage capacitor(s) have reached 84.1 percent of the peak voltage of the power supply. The indicator light shall be mounted on the rear of the light unit behind a prismatic or jeweled lens and shall be readily visible 95 degrees each side of the axis of the light unit rear.

3.16.6 Triggering Circuit. - The triggering circuit of the flash unit shall be so designed that the light may be flashed by means of a phototube or wire. The triggering circuit shall be a low energy type, shall not fire due to impulses generated by a battery capacitor solenoid circuit and shall not damage the synchronization contacts of camera shutters when the light is flashed.

3.16.7 Synchronization Connector. - The light unit shall be provided with a parallel blade polarized type female receptacle to accept any standard, commercial, polarized type male synchronization cable connector. The connector shall be marked "SYNC". Proper polarity shall be maintained between the flash unit, the camera shutter contacts, and the camera body.

3.16.8 Mounting. - The light unit shall be provided with a tripod socket, a mounting collar, and a camera accessory foot.

3.16.8.1 Tripod Socket. - A tripod socket, 1/4-20, in accordance with American Standard PH3.6 shall be provided.

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3.16.8.2 Mounting Collar. - The mounting collar shall be capable of being securely mounted and properly indexed on any standard 1-1/2 inch outside diameter flash gun battery case. The collar may be either metal or rubber. The collar shall be fastened to the light unit tripod socket by means of a captive thumbscrew. If a metal collar is provided, a captive thumbscrew shall be provided for securing the collar to the flash gun battery case.

3.16.8.3 Camera Accessory Foot. - A camera accessory foot shall be provided, fabricated in accordance with the requirements of American Standard PH3.30. The foot shall be interchangeable with the mounting collar. The height of the foot shall be such that there is no interference with the operation or functions of any camera by the light unit.

3.17 ELECTRICAL CIRCUIT. - The electrical circuit shall permit operation of the flash unit from either an external power source of 115 plus or minus 5 volts, 50-60 cycles, single phase, alternating current or from the battery supplied as part of the power supply. Battery (BAT), AC and the battery charging (CHG) operations shall be independent and the equipment shall meet all the specified performance requirements in any mode of operation.

3.17.1 High Voltage Circuit. - The high voltage circuit shall have a nominal voltage of 450 volts direct current (vdc) with a maximum peak voltage of 500 vdc for the Type I power supply. The Type II power supply shall have a nominal voltage of 900 vdc with a maximum peak voltage of 1000 vdc. The peak voltage shall be measured in accordance with the procedures described in American Standard PH3.40.

3.17.2 Battery Charging Circuit. - The battery charging circuit shall be a constant current type and shall meet the battery charging requirements specified herein.

3.17.3 Transformer(s). - The transformer(s) shall be heavy duty types designed for the intended purpose and shall meet the requirements specified herein. The transformer(s) may be encapsulated.

3.17.4 Rectifier(s). - Hermetically sealed silicon rectifiers shall be used.

3.17.5 High Voltage Capacitor(s). - The capacitor(s) shall be low leakage type and shall be encased in either metal or plastic.

3.17.6 Current Oscillator(s). - The current oscillator(s) may be either vibrator(s), transistor oscillator(s), or any suitable current oscillating components.

3.18 SAFETY MEASURES. - Safety measures shall be provided against high voltages, short circuits or power overloading for the protection of the equipment and to operating personnel.

3.18.1 Interlocks. - Interlocks shall be provided where necessary to

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prevent access to the interior of the equipment, or compartments thereof, without the removal of all voltages exceeding 150 volts above ground potential. In lieu of interlocks, suitable high voltage insulating barriers may be used provided that the barriers are suitable and permanently marked or labeled with a warning that all voltages are to be removed from the unit before removing the barrier.

3.18.2 Fuses. - Fuses shall be provided in the AC and DC parts of the power supply electrical circuitry. The fuses shall be so positioned that they are readily accessible and may be replaced without the use of tools. The fuses shall be the slow-blow type.

3.18.3 Dummy Plug. - One dummy plug shall be provided for insertion into the unused light unit connector of the Type II power supply. The plug shall be attached to the power supply by any suitable means to prevent its loss.

3.18.4 Warning Plate. - A suitable warning plate, permanently and legibly inscribed "HIGH VOLTAGE" shall be permanently mounted in a conspicuous place within the power supply. The plate shall state that high voltages are present and the necessary precautions be taken to remove the high voltages present before disassembly for maintenance or repair.

3.18.5 Warning Label. - A warning label, with before-use instructions printed thereon, shall be affixed to the power supply in a conspicuous place. The instructions shall be printed or stamped on a self adhesive label. Removal of the label shall not damage the finish of the equipment. The instructions to be printed or stamped on the label are as follows:

BEFORE USE INSTRUCTIONS

THE BATTERY HAS BEEN SHIPPED IN A DISCHARGED STATE. BEFORE USING THE FLASH UNIT, CHARGE THE BATTERY ACCORDING TO THE BATTERY CHARGING INSTRUCTIONS IN THE OPERATOR'S MANUAL. PLACE THE UNIT ON AC OPERATION FOR AT LEAST ONE HOUR WITHOUT FLASHING THE UNIT AFTER THE BATTERY HAS BEEN CHARGED. AFTER THE FOREGOING INSTRUCTIONS HAVE BEEN COMPLIED WITH, THE FLASH UNIT IS READY FOR USE.

3.19 EXTERNAL POWER CABLE. - An external power cable 15 feet plus or minus 6 inches in length shall be provided with a female connector to mate with the male connector on the power supply on one end and with a parallel bladed connector on the other end. The parallel bladed connector may be either a commercial type or Type UP120M conforming to Military Standard MS91184. The cable conductors shall be capable of carrying a minimum of 230 volts of electrical current with complete safety to the equipment and operating personnel.

3.20 COLOR. - The external color of the power supply and light unit shall be a medium gray unless otherwise specified in the contract or order (see 6.2).

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3.21 FINISH. - Each flash unit shall be given a protective finish, if applicable, in accordance with Specification MIL-P-14072. This shall include finish of hardware type of items, such as handles, hinges, screws, etc., and necessary touch-up after mounting.

3.22 MARKING OF PARTS. - All part identification marking shall be in accordance with the requirements of Military Standard MIL-STD-130. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the items in which they are used.

3.23 IDENTIFICATION PLATE. - An identification plate, conforming to the requirements of Military Standard MIL-STD-130 shall be permanently attached to the outside of the power supply.

3.24 SERIAL NUMBER. - A serial number shall be assigned by the manufacturer to each flash unit. This number shall not be duplicated in any one contract or order.

3.25 TECHNICAL LITERATURE. - When specified in the contract or order (see 6.2), one instruction manual covering the operation, maintenance and repair of the flash unit shall be furnished with each flash unit.

3.26 SPARE PARTS. - Spare parts shall be furnished as specified in the contract or order (see 6.2). Spare parts shall be identical to corresponding parts of the flash unit furnished on the contract or order.

3.27 DRAWINGS. - When specified in the procurement document, (see 6.2), a complete set of drawings and all associated lists conforming to the requirements of Military Specification MIL-D-70327 shall be provided.

3.28 MICROFILM OF DRAWINGS AND RELATED DATA. - When specified in the procurement document, (see 6.2), microfilm of the drawings and associated lists conforming to the requirements of Military Specification MIL-M-9368 shall be provided.

3.29 WORKMANSHIP. - The flash unit shall be manufactured and assembled in accordance with the best commercial practice in the manufacture and assembly of electronic equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION. - Unless otherwise specified in the contract or order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, (see 6.2), the supplier may utilize his own facilities

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or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 TEST CONDITIONS AND MATERIAL. - Unless otherwise specified herein, tests shall be conducted under ordinary room conditions of temperature and humidity. The equipment and supplies necessary for testing shall be furnished by the contractor.

4.3 CLASSIFICATION OF INSPECTIONS. - The examination and testing of flash units shall be classified as follows:

(a) Preproduction Inspection

(b) Quality Conformance Inspection

(c) Preparation for Delivery Inspection

4.4 PREPRODUCTION INSPECTION. - Prior to submission for acceptance, the preproduction sample shall be subjected by the contractor to the preproduction inspection specified herein. When specified (see 6.2) a representative of the requiring activity shall monitor any, or all, of the preproduction tests conducted by the contractor. On completion of the inspection by the contractor, the preproduction sample shall be shipped to the Military Testing Agency designated by the requiring activity (see 6.2). Unless otherwise specified (see 6.2) a report of the results of the tests shall accompany the preproduction sample. The Military Testing Agency may repeat the preproduction inspection or make other tests as needed to verify that the production methods and choice of design details are capable of yielding equipment that complies with the requirements of this specification. Approval by the requiring activity shall be based on no defects in the preproduction sample. The preproduction inspection shall consist of all the examinations and tests specified herein.

4.5 QUALITY CONFORMANCE INSPECTION. - Each production flash unit, subsequent to the preproduction sample, submitted for acceptance under the contract or order, shall be subjected by the contractor to the quality conformance inspection specified herein. The quality conformance inspection shall verify that all subsequent flash unit are duplicates of the approved preproduction sample. The quality conformance inspection shall consist of the following examinations and tests as specified herein.

(a) Visual and Mechanical Examination, paragraph 4.10.1

(b) Interchangeability Examination, paragraph 4.10.2

(c) Light Output Test, paragraph 4.10.10

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(d) Reflector Coverage Test, paragraph 4.10.11

(e) Recycle Test, paragraph 4.10.12

(f) Phototube Operation Test, paragraph 4.10.15

(g) Component Reliability Test, paragraph 4.10.17

4.6 PREPARATION FOR DELIVERY INSPECTION. - Examine the preservation, packaging, packing and markings for shipment and storage, to determine conformance to the requirements of the applicable specifications referenced herein.

4.7 SAMPLING. - Sampling procedures to determine compliance with the specified quality assurance provisions shall be in accordance with Military Standard MIL-STD-105. Unless otherwise specified herein, the inspection level shall be General Inspection Level II and the acceptable quality lever (AQL) shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.8 LOT. - For the purposes of sampling, a lot shall consist of all the flash units of the same type, offered for acceptance at one time.

4.9 CLASSIFICATION OF DEFECTS. - Classification of defects shall be as defined in Military Standard MIL-STD-105.

4.10 EXAMINATIONS AND TESTS. -

4.10.1 Visual and Mechanical Examination. - Examine all the components of the flash unit visually and with gauges to determine conformance with the requirements of this specification with respect to design, size, weight, materials, construction, fit of components, finish, color, markings, completeness, and workmanship. Operate all controls and fasteners to determine that they function properly.

4.10.2 Interchangeability Examination. - Like units, components, assemblies, sub-assemblies, and replaceable parts shall be examined or physically interchanged to determine compliance with the interchangeability requirements of this specification. Non-interchangeability shall be considered a major defect.

4.10.3 Extreme Temperature Cycle Test. - The flash unit shall be subjected to the temperature cycles shown in Military Standard MIL-STD-169 except as modified below:

Step	Requirements
1	As specified
2	As specified
3	Delete the requirement
4	125 plus 5 minus 0 degrees F
5	As specified
6	Delete the requirement
7	As specified
8	Delete the requirement
9	0 plus 0 minus 5 degrees F
10	As specified

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Exposure at the high temperature extreme, step 2 above, shall not exceed 4 hours and at the low temperature, step 7 above, shall not exceed 24 hours at any time. Measurements shall be taken with the flash unit being operated on BAT and AC.

Step	Measurement	Test Paragraph	Performance Requirement
1, 4, 5, 9 and 10	Light Output	4.10.10	As Specified in 3.11.1
1, 4, 5, 9 and 10	Recycle Time	4.10.12	As Specified in 3.11.1

The light output measurement shall be taken on the principle axis of the lamphead only and only the probe of the integrating light meter shall be placed inside the temperature chamber. Failure of the flash unit to meet the requirements of the above test shall be cause for rejection.

4.10.4 Moisture Resistance Test. - The flash unit shall be subjected to the Humidity Test of MIL-T-5422. The flash unit shall be flashed 5 times each on AC and BAT at the end of each complete humidity test cycle.

4.10.5 Vibration Test. - Verify that the flash unit can be flashed on both AC and BAT and that the charging circuit operates satisfactorily. Secure the equipment directly to a vibration table. Adjust the vibration table to provide a total excursion of 0.050 inch plus or minus 10 percent. The frequency shall be varied uniformly between the approximate limits of 10 and 55 cycles per second (cps). The entire frequency range from 10 to 55 cps and return to 10 cps shall be traversed in approximately 15 minutes. This motion shall be applied for a period of one hour in each of 3 mutually perpendicular directions. At the end of the test on each axis, the flash unit shall be flashed on both AC and BAT. At the conclusion of the test the battery charging circuit shall operate satisfactorily and the equipment shall be examined to determine that no damage has been caused by the test.

4.10.6 Impact Test. - The flash unit shall be subjected to Method 205C, Test Condition C of Military Standard MIL-STD-202. In addition, each corner shall be subjected once to the same conditions of impact.

4.10.7 Radio Interference Suppression Test. - Unless otherwise specified in the contract or order (see 6.2), the flash unit shall be tested in accordance with the procedures specified in MIL-I-16910 to determine conformance to the Radio Interference Suppression requirements specified in Section 3.

4.10.8 Elevation Test. - The flash unit shall be placed in an altitude chamber and tested as follows:

(a) With the flash unit placed in the chamber, operating on BAT and stabilized at room temperature, measure the recycle time.

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(b) Reduce the barometric pressure to 20.6 plus or minus 0.1 inches of mercury (simulated 10,000 feet above sea level), and after temperature and pressure stabilization, measure the recycle time. The flash unit shall meet the recycle time specified herein.

(c) Further reduce the barometric pressure to 3.4 plus or minus 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Measure the recycle time. The flash unit shall meet the recycle time specified herein.

4.10.9 Operating Life. - The flash unit shall be tested as follows:

(a) Install a new, unused, and discharged battery in the test flash unit power supply.

(b) Charge the battery for 14 hours.

(c) Flash the test flash unit using battery power 100 times for the Type I Flash Unit and 200 times for the Type II Flash Unit.

(d) Recharge the battery for 14 hours.

(e) Repeat step (c) above.

(f) Turn the selector switch to AC and flash the test flash unit using an external ac power source of 115 plus or minus 2 volts, 50-60 cycles, single phase, until a total of 10,000 flashes have been made.

Flashing shall occur at a minimum of 84.1 percent of peak voltage of the test flash unit. The first 500 flashes shall be performed with the flash unit connected to a camera shutter by means of a synchronization cable and the test flash unit flashed by manually or mechanically tripping the shutter release lever of the camera shutter.

4.10.10 Light Output Test. - The light output shall be measured in accordance with the procedures described in American Standard PH3.40. The light output shall be a minimum of 1,200 Beam Candle Power Seconds (BCPS) for the Type I Flash Unit and 2,400 BCPS for the Type II Flash Unit. The meter used to measure the light output shall be an integrating type of light meter such as the General Radio Company Type 1501-A Integrating Light Meter. The meter shall have been calibrated not more than 30 days prior to the test.

4.10.11 Reflector Coverage Test. - The reflector coverage of the light unit shall be measured in accordance with the procedures described in American Standard PH3.40. The test may be performed simultaneously with the light output test. The meter used for this test shall be the same meter used for the light output test.

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4.10.12 Recycle Test. - The recycle test shall be measured in accordance with the procedures described in American Standard PH3.40. The test shall be made during BAT and AC operation of the flash unit. The recycle time shall not exceed 10 seconds for either mode of operation.

4.10.13 Synchronization Test. - The flash unit shall be tested for synchronization in accordance with the procedures described in American Standard PH3.40 for use with cameras equipped with X type synchronization contacts for flash photography.

4.10.14 Flash Duration Test. - The flash duration of the flash unit shall be measured in accordance with the procedures described in American Standard PH3.40. The flash duration shall not be shorter than 1/2500 second. The test shall be made during both BAT and AC operation.

4.10.15 Phototube Operation Test. - The flash unit, with a suitable phototube installed, shall be flashed by means of the light emitted from a Type I flash unit, or equivalent, at a distance of 50 feet under not more than 50 foot candles of ambient light. One flash unit from the sample shall be used for this test.

4.10.16 Battery Charging Circuit Test. - A fully discharged battery shall be placed in the power supply of the flash unit power supply under test. The flash unit shall be connected to a 115 plus or minus 2 volts, 50-60 cycles, single phase ac external power source. Set the selector switch of the flash unit to CHG. Measure the time required for the battery to reach full charge. The battery shall reach full charge in 14 hours or less. Continue the charging for a total of 24 hours and note the charge and battery condition. The battery shall not have been adversely affected by this overcharging period and shall pass the number of flashes per battery charge test.

4.10.17 Component Reliability Test. - The flash unit, with a fully charged battery installed in the power supply, shall be flashed every 15 seconds with the selector switch set on BAT. The number of flashes shall be 100 and 200 flashes for the Type I and Type II Flash Units respectively. Recharge the battery for 14 hours. Repeat the flashing cycle. Allow the test unit to stand without use for 24 hours. The test unit shall then be operated on AC from an external power source of 115 plus or minus 2 volts, 50-60 cycles, ac power. The test unit shall then be flashed for an additional 500 times at 15 second intervals. The test unit shall comply with the requirements of this specification without replacement or adjustment of any part or component. One flash unit from the sample shall be used for this test.

4.10.18 Battery Useful Life Test. - The battery shall be subjected to 100 cycles of charge and discharge. The battery at the completion of this test shall conform to the requirements specified and shall not have declined below 95 percent of its nominal capacity. In lieu of this test, the contractor may furnish a Certificate of Compliance that the battery meets the useful life requirements as specified.

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4.10.19 Number of Flashes per Battery Charge. - Install a fully charged battery into the power supply. Measure the voltage of the battery. Flash the unit every 15 seconds, counting the flashes, until the battery voltage is 4.2 volts dc. The number of flashes shall not be less than specified. One flash unit from the sample shall be used for this test.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING. - Preservation and packaging shall be Level A as defined in Federal Standard No. 102. The flash unit consisting of the power supply, battery, external power cable, light unit(s), and light unit mounting accessories shall be packaged as one unit. The battery shall be in a discharged state and shall be installed in the power supply but not connected. The battery leads shall be suitably protected to prevent accidental contact with metal surfaces.

5.1.1 Cleaning. - The flash unit shall be cleaned following the procedures specified in Specification MIL-P-116, for C-1 cleaning.

5.1.2 Drying. - Cleaned surfaces shall be dried in accordance with the procedures specified in Specification MIL-P-116, for D-4 drying.

5.1.3 Preservation. - Preservatives shall not be used on the flash unit.

5.1.4 Unit Package. - The flash unit shall be packaged Method IIb as prescribed in Specification MIL-P-116.

5.2 PACKING. - The level of packing to be used shall be as specified by the requiring activity (see 6.2).

5.2.1 Level A and C. - The flash unit, preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with the requirements of Specification MIL-P-7936.

5.3 MARKING. - All unit packages and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 INTENDED USE. - Flash Units, Photographic, Repeating, LE-25( ) and LE-26( ) are for use as main or supplementary light sources for synchronized flash photography with any camera having X type synchronization contacts.

6.2 ORDERING DATA. - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Quantity desired.

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(c) Type of unit required (see 1.2).

(d) Number of light units required for the Type II Flash Unit (see 3.16).

(e) Data, if required (see 3.2, 3.25, 3.27, 3.28, and 4.4).

(f) Preproduction sample (see 3.4 and 6.3).

(g) Radio Interference Suppression, if not as specified (see 3.13 and 4.10.7).

(h) Batteries, if not as specified (see 3.15.1).

(i) Color of equipment (see 3.14.2 and 3.20).

(j) Commercial or AN connectors (see 3.14.6).

(k) Spare parts, if required (see 3.26).

(l) Responsibility for inspection (see 4.1).

(m) Monitoring of tests (see 4.4).

(n) Designation of Military Testing Agency (see 4.4).

(o) Level of packing for shipment (see 5.2).

6.3 PREPRODUCTION. - If the flash unit has passed the preproduction inspection during the past twelve months, the requirements for a preproduction sample of additional flash units may be waived. Authorization to omit the preproduction sample may be requested by the contractor from the requiring activity.

6.4 TYPE DESIGNATOR. - In accordance with the provisions of Military Standard MIL-STD-155, the parenthesis in the type designator will be replaced by a letter, number, or both. This information will be furnished upon application by the contractor to the requiring activity after award of each contract or order. The complete type designator shall then be used on nameplates, in instruction books, and wherever reference is made to the equipment covered by this specification.

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Custodians:

Army - EL
Navy - WP
Air Force - 67

Review activities

Navy - WP, MC
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User activities
Army - EL, MO

Preparing activity
Navy - WP

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

M11-P-81226

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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